

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular Winter Examination – 2024

Course: B.Tech Branch : Electronics and Computer Engineering/ Electronics and Computer Science

Semester : VII

Subject Code & Name: BTECPC701 Industry 4.0 and Automation

Max Marks: 60

Date: 05/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	What is an IoT?		1				
	<table border="1"> <tr> <td>a. network of virtual objects</td> <td>b. network of objects in the ring structure</td> <td>c. network of physical objects embedded with sensors</td> <td>d. network of sensors</td> </tr> </table>	a. network of virtual objects	b. network of objects in the ring structure	c. network of physical objects embedded with sensors	d. network of sensors		
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2	Which layer is used for wireless connection in IoT devices?		1				
	<table border="1"> <tr> <td>a. Application layer</td> <td>b. Network layer</td> <td>c. Data link layer</td> <td>d. Transport layer</td> </tr> </table>	a. Application layer	b. Network layer	c. Data link layer	d. Transport layer		
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3	What is Cloud Computing?		1				
	<table border="1"> <tr> <td>a. Cloud Computing means storing data in a database</td> <td>b. Cloud Computing is a tool used to create an application</td> <td>c. Cloud Computing means providing services like storage, servers, database, networking, etc</td> <td>d. None of these</td> </tr> </table>	a. Cloud Computing means storing data in a database	b. Cloud Computing is a tool used to create an application	c. Cloud Computing means providing services like storage, servers, database, networking, etc	d. None of these		
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4	The core element of architecture of smart city is _____		1				

	a. Integrated Information Provider	b. Management center	c. Urban Application Platform	d. Mobile Unified Service		
5	Which of the following is a type of cloud computing service?					1
	a. Service-as-a-Software (SaaS)	b. Software-as-a-Service (SaaS)	c. Software-and-a-Server (SaaS)	d. Software-as-a-Server (SaaS)		
6	A technology for application of electronics, mechanical and computer based system to control and operate the system is called,					1
	a. PLC	b. Automation	c. Sequential controller	d. Microprocessor based system		
7	Which are the applications of cloud robotics?					1
	a. Cloud medical robots	b. Autonomous mobile robots	c. Assistive robots	d. All of these		
8	The collaborative robot arms are designed to mimic the range of motion of a _____					1
	a. Human arm	b. Network	c. Device	d. Machine arm		
9	The vision of Industry 4.0 is					1
	a. More efficient use of natural resources and energy	b. To decrease the cost of industrial production	c. Enabling a custom mass production without significantly increasing overall production costs	d. All of these		
10	Which of the following is generally considered to be a key enabler of the Third Industrial Revolution?					1
	a. Robot	b. Personal Computer	c. Programmable	d. Printer		

			Logic Controller (PLC)			
11	Which of the following is the central processing unit of PLC?					1
	a. Processor	b. Power supply	c. Memory unit	d. I/O module		
12	The SCADA stands for,					1
	a. Supervisory Column and Data Assessment	b. Supervisory Column and Data Assessment	c. Supervisory Control and Document Acquisition	d. Supervisory Control and Data Acquisition		
Q. 2	Solve the following.					12
A)	Discuss the historical context of fourth revolution in industry (Industry 4.0)					6
B)	What is the cloud computing? Explain the service models of cloud computing.					6
Q.3	Solve the following.					12
A)	Explain the different building blocks of IIoT Business Model and its advantages.					6
B)	Explain with suitable example the applications of Internet of things (IOT) in various domains.					6
Q. 4	Solve Any Two of the following.					12
A)	Explain the Cognitive Architecture for Cyber Physical Robotics					6
B)	Explain the advance sensor technologies used in Robotics					6
C)	Explain in detail the different types and technological components of robots.					6
Q.5	Solve Any Two of the following.					12
A)	Explain the architecture of Industrial Automation Systems and give its advantages.					6
B)	Explain the types of Digital Twins and give its benefit, impact, and features.					6
C)	Discuss the different applications of Digital Twin in production, manufacturing field.					6

Q. 6	Solve Any Two of the following.		12
A)	Explain with suitable diagram components of SCADA system.		6
B)	Give the comparison in between PLC, DCS and SCADA system.		6
C)	Write short note on Human Machine Interface (HMI).		6
*** End ***			